

```
R1(config)#
*Aug 27 01:47:01.220: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#ip route 32.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 45.0.0.0 255.0.0.0 12.0.0.2
R1(config)#ip route 172.36.0.0 255.255.0.0 12.0.0.2
R1(config)#ip route 223.0.50.0 255.255.255.0 12.0.0.2
R1(config)#exit
```

```
R1#
*Aug 27 01:48:56.566: %SYS-5-CONFIG_I: Configured from console by console
```

```
R1#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	192.50.20.1	YES	manual	up	up
Ethernet0/1	12.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/1
L    12.0.0.1/32 is directly connected, Ethernet0/1
S    23.0.0.0/8 [1/0] via 12.0.0.2
S    24.0.0.0/8 [1/0] via 12.0.0.2
S    26.0.0.0/8 [1/0] via 12.0.0.2
S    32.0.0.0/8 [1/0] via 12.0.0.2
S    45.0.0.0/8 [1/0] via 12.0.0.2
S    172.36.0.0/16 [1/0] via 12.0.0.2
192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.50.20.0/24 is directly connected, Ethernet0/0
L    192.50.20.1/32 is directly connected, Ethernet0/0
S    223.0.50.0/24 [1/0] via 12.0.0.2
```

```
R1#
R1#
```



```
R2#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	12.0.0.2	YES	manual	up	up
Ethernet0/1	26.0.0.1	YES	manual	up	up
Ethernet0/2	23.0.0.1	YES	manual	up	up
Ethernet0/3	24.0.0.1	YES	manual	up	up

```
R2#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from Pfr

```
Gateway of last resort is not set
```

```

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/0
L    12.0.0.2/32 is directly connected, Ethernet0/0
23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/2
L    23.0.0.1/32 is directly connected, Ethernet0/2
24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    24.0.0.0/8 is directly connected, Ethernet0/3
L    24.0.0.1/32 is directly connected, Ethernet0/3
26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.1/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 23.0.0.2
S    45.0.0.0/8 [1/0] via 24.0.0.2
S    172.36.0.0/16 [1/0] via 23.0.0.2
S    192.50.20.0/24 [1/0] via 12.0.0.1
S    223.0.50.0/24 [1/0] via 26.0.0.2
R2#
```



30°C
Mostly sunny



Search



144Hz / Adaptive SYNC


```
R3(config)#do sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	23.0.0.2	YES	manual	up	up
Ethernet0/1	32.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R3(config)#do sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

```
Gateway of last resort is not set
```

```
S    12.0.0.0/8 [1/0] via 23.0.0.1
    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/0
L    23.0.0.2/32 is directly connected, Ethernet0/0
S    24.0.0.0/8 [1/0] via 23.0.0.1
S    26.0.0.0/8 [1/0] via 23.0.0.1
    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.1/32 is directly connected, Ethernet0/1
S    45.0.0.0/8 [1/0] via 23.0.0.1
S    172.36.0.0/16 [1/0] via 32.0.0.2
S    192.50.20.0/24 [1/0] via 23.0.0.1
S    223.0.50.0/24 [1/0] via 23.0.0.1
```

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R3(config)#
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R3(config)#
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R3(config)#
```



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Mostly sunny



Q Search



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```
R4#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	24.0.0.2	YES	manual	up	up
Ethernet0/1	45.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R4#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

```
S    12.0.0.0/8 [1/0] via 24.0.0.1
S    23.0.0.0/8 [1/0] via 24.0.0.1
     24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      24.0.0.0/8 is directly connected, Ethernet0/0
L      24.0.0.2/32 is directly connected, Ethernet0/0
S    26.0.0.0/8 [1/0] via 24.0.0.1
S    32.0.0.0/8 [1/0] via 45.0.0.2
     45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      45.0.0.0/8 is directly connected, Ethernet0/1
L      45.0.0.1/32 is directly connected, Ethernet0/1
S    172.36.0.0/16 [1/0] via 45.0.0.2
S    192.50.20.0/24 [1/0] via 24.0.0.1
S    223.0.50.0/24 [1/0] via 24.0.0.1
R4#
```



```
R5(config)#do sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	172.36.0.1	YES	manual	up	up
Ethernet0/1	32.0.0.2	YES	manual	up	up
Ethernet0/2	45.0.0.2	YES	manual	up	up
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R5(config)#do sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S 12.0.0.0/8 [1/0] via 45.0.0.1
S 23.0.0.0/8 [1/0] via 32.0.0.1
S 24.0.0.0/8 [1/0] via 45.0.0.1
S 26.0.0.0/8 [1/0] via 45.0.0.1
32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 32.0.0.0/8 is directly connected, Ethernet0/1
L 32.0.0.2/32 is directly connected, Ethernet0/1
45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 45.0.0.0/8 is directly connected, Ethernet0/2
L 45.0.0.2/32 is directly connected, Ethernet0/2
172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.36.0.0/16 is directly connected, Ethernet0/0
L 172.36.0.1/32 is directly connected, Ethernet0/0
S 192.50.20.0/24 [1/0] via 45.0.0.1
S 223.0.50.0/24 [1/0] via 45.0.0.1
```

```
R5(config)#
```



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Mostly sunny



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```
R6(config)#
```

```
R6(config)#do sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	223.0.50.1	YES	manual	up	up
Ethernet0/1	26.0.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R6(config)#do sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    12.0.0.0/8 [1/0] via 26.0.0.1
S    23.0.0.0/8 [1/0] via 26.0.0.1
S    24.0.0.0/8 [1/0] via 26.0.0.1
    26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.2/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 26.0.0.1
S    45.0.0.0/8 [1/0] via 26.0.0.1
S    172.36.0.0/16 [1/0] via 26.0.0.1
S    192.50.20.0/24 [1/0] via 26.0.0.1
    223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    223.0.50.0/24 is directly connected; Ethernet0/0
L    223.0.50.1/32 is directly connected, Ethernet0/0
```

```
R6(config)#
```

```
R6(config)#
```

```
R6(config)#
```


QEMU (Win-1)

C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address : 192.50.20.100
Subnet Mask : 255.255.255.0
Default Gateway : 192.50.20.1

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=3ms TTL=124
Reply from 172.36.0.10: bytes=32 time=3ms TTL=124
Reply from 172.36.0.10: bytes=32 time=3ms TTL=124
Reply from 172.36.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 172.36.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 3ms, Average = 3ms

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=2ms TTL=125
Reply from 223.0.50.10: bytes=32 time=2ms TTL=125
Reply from 223.0.50.10: bytes=32 time=3ms TTL=125
Reply from 223.0.50.10: bytes=32 time=2ms TTL=125

Ping statistics for 223.0.50.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

Start

C:\WINDOWS\system...

223.0.50.10

Win-2

C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

```
Connection-specific DNS Suffix . :  
IP Address. . . . . : 223.0.50.10  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 223.0.50.1
```

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

```
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128  
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128  
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128  
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
```

Ping statistics for 223.0.50.10:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

```
Reply from 172.36.0.10: bytes=32 time=4ms TTL=124  
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124  
Reply from 172.36.0.10: bytes=32 time=1ms TTL=124  
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
```

Ping statistics for 172.36.0.10:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 1ms, Maximum = 4ms, Average = 2ms
```

C:\Documents and Settings\user>ping 192.50.20.100

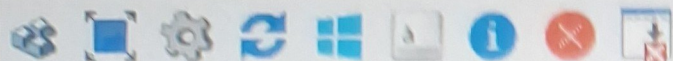
Pinging 192.50.20.100 with 32 bytes of data:

```
Reply from 192.50.20.100: bytes=32 time=1ms TTL=125  
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125  
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125  
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
```

Ping statistics for 192.50.20.100:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

C:\Documents and Settings\user>



C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

```
Connection-specific DNS Suffix . :  
IP Address. . . . . : 172.36.0.10  
Subnet Mask . . . . . : 255.255.0.0  
Default Gateway . . . . . : 172.36.0.1
```

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

```
Reply from 223.0.50.10: bytes=32 time=4ms TTL=124  
Reply from 223.0.50.10: bytes=32 time=3ms TTL=124  
Reply from 223.0.50.10: bytes=32 time=3ms TTL=124  
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
```

Ping statistics for 223.0.50.10:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 2ms, Maximum = 4ms, Average = 3ms
```

C:\Documents and Settings\user>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

```
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124  
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124  
Reply from 192.50.20.100: bytes=32 time=3ms TTL=124  
Reply from 192.50.20.100: bytes=32 time=3ms TTL=124
```

Ping statistics for 192.50.20.100:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

C:\Documents and Settings\user>_